

Mini Assessment

Smart API Reliability & JSON Data Validation

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Questions & Answers

1. If an API returns 500, what should your program do?

Answer: HTTP 500 means an Internal Server Error. The program should not assume the request succeeded. It should record or report the error, avoid repeated immediate requests, and retry only when appropriate. For temporary server problems, a limited retry with a short backoff can be used.

2. If an API request times out, what should your program do?

Answer: A timeout means the server did not respond within the allowed time. The program should catch the timeout error and handle it safely instead of crashing. It can retry a limited number of times with a delay and show a clear message if the request still fails.

3. If a JSON object may not contain a key, why is .get() useful?

Answer: Using `data["key"]` raises a `KeyError` when the key is missing. Using `data.get("key")` safely returns `None` when the key is absent, or a default value can be supplied. For example:
`data.get("status", "Not Available")`.

Quick Revision

Status / Situation	What to do
500 Internal Server Error	Handle the error safely; retry only when appropriate with limited backoff.
Request timeout	Catch the timeout, retry a limited number of times, then show a clear error message.
JSON key may be missing	Prefer <code>.get("key")</code> or <code>.get("key", default)</code> to avoid <code>KeyError</code> .